Mechanical Properties of Solids and Fluids

Single Correct Type Questions

1. The Young's modulus of a steel wire of length 6 *m* and cross-sectional area 3 mm^2, is $2\times11^{11}\ N/m^2$. The wire is suspended from its support on a given planet. A block of mass 4 *kg* is attached to the free end of the wire. The acceleration due to gravity on the planet is $\frac{1}{4}$ of its value on the earth. The elongation of wire is (Take *g* on the earth = 10 m/s^2) **[1 Feb, 2023 (Shift-II)]**

(*a*) 1 *cm* (*b*) 1 *mm*
(*c*) 0.1 *mm* (*d*) 0.1 *cm*

2. A force is applied to a steel wire 'A', rigidly clamped at one end. As a result elongation in the wire is 0.2 mm. If same force is applied to another steel wire 'B' of double the length and a diameter 2.4 times that of the wire 'A', the elongation in the wire 'B' will be (wires having uniform circular cross sections) **[30 Jan, 2023 (Shift-II)]**

(*a*) 6.06×10^{-2} mm
(*b*) 2.77×10^{-2} mm
(*c*) 3.0×10^{-2} mm
(*d*) 6.9×10^{-2} mm

3. A wire of length L is hanging from a fixed support. The length changes to L_1 and L_2 when masses 1kg and 2kg are suspended respectively from its free end. Then the value of L is equal to: **[29 June, 2022 (Shift-I)]**

(*a*) $\sqrt{L_1L_2}$ (*b*) $\frac{L_1+L_2}{2}$ (*c*) $2L_1-L_2$ (*d*) $3L_1-2L_2$

4. The length of a metal wire is ℓ_1, when the tension in it is T_1 and is ℓ_2 when the tension is T_2. The natural length of the wire is: **[20 July, 2021 (Shift-II)]**

(*a*) $\sqrt{\ell_1\ell_2}$

(*b*) $\frac{\ell_1+\ell_2}{2}$

(*c*) $\frac{\ell_1T_2-\ell_2T_1}{T_2-T_1}$

(*d*) $\frac{\ell_1T_2+\ell_2T_1}{T_2+T_1}$

5. Young's moduli of two wires A and B in the ratio 7 : 4. Wire A is 2 m long and has radius R. Wire B is 1.5 m long and has radius 2 mm. If the two wires stretch by the same length for a given load, then the value of R is close to **[8 April, 2019 (Shift-II)]**

(*a*) 1.9 mm (*b*) 1.7 mm
(*c*) 1.5 mm (*d*) 1.3 mm

6. A uniform cylindrical rod of length L and radius r, is made from a material whose Young's modulus of Elasticity equals Y. When this rod is heated by temperature T and simultaneously subjected to a net longitudinal compressional force F, its length remains unchanged. The coefficient of volume expansion, of the material of the rod, is (nearly) equals to: **[12 April, 2019 (Shift-II)]**

(*a*) $9F/(\pi r^2YT)$
(*b*) $F/(3\pi r^2YT)$
(*c*) $3F/(\pi r^2YT)$
(*d*) $6F/(\pi r^2YT)$

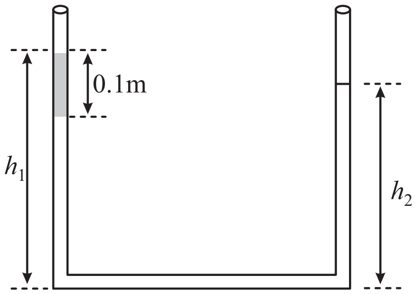
7. The normal density of a material its ρ and its bulk modulus of elasticity is K. The magnitude of increase in density of material, when a pressure P is applied uniformly on all sides, will be : **[26 Feb, 2021 (Shift-I)]**

(*a*) $\frac{K}{\rho P}$ (*b*) $\frac{PK}{\rho}$
(*c*) $\frac{\rho K}{P}$ (*d*) $\frac{\rho P}{K}$

8. Two steel wires having same length are suspended from a ceiling under the same load. If the ratio of their energy stored per unit volume is 1 : 4, the ratio of their diameters is: **[9 Jan, 2020 (Shift-II)]**

(*a*) 1 : 2 (*b*) $1:\sqrt{2}$
(*c*) 2 : 1 (*d*) $\sqrt{2}:1$

9. An open-ended U-tube of uniform cross-sectional area contains water (density 10^3 kg m^{-3}).Initially the water level stands at 0.29 m from the bottom in each arm. Kerosene oil (a water-immiscible liquid) of density 800 kg m^{-3} is added to the left arm until its length is 0.1 m, as shown in the schematic figure below. The ratio $\left(\frac{h_1}{h_2}\right)$ of the heights of the liquid in the two arms is:



[JEE Adv, 2020]

(*a*) $\frac{15}{14}$ (*b*) $\frac{35}{33}$ (*c*) $\frac{7}{6}$ (*d*) $\frac{5}{4}$

10. A small spherical droplet of density d is floating exactly half immerse in a liquid ρ and surface tension T. The radius of the droplet is (take note that the surface tension applies an upward force on the droplet): **[9 Jan, 2020 (Shift-II)]**

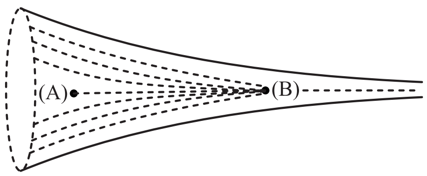
(*a*) $r=\sqrt{\frac{T}{(d+\rho)g}}$

(*b*) $r=\sqrt{\frac{3T}{(2d-\rho)g}}$

(*c*) $r=\sqrt{\frac{T}{(d-\rho)g}}$

(*d*) $r=\sqrt{\frac{2T}{3(d-\rho)g}}$

11.



The figure shows a liquid of given density flowing steadily in horizontal tube of varying cross-section. Cross sectional areas at A is 1.5 cm^2 and B is 25 mm^2, if the speed of liquid at B is 60 cm/s then $(P_A - P_B)$ is:

(Given P_A and P_B are liquid pressures at A and B points.

Density ρ = 1000 kg m^{-3} A and B are on the axis of tube

[13 April, 2023 (Shift-I)]

(*a*) 175 Pa (*b*) 27 Pa

(*c*) 135 Pa (*d*) 36 Pa

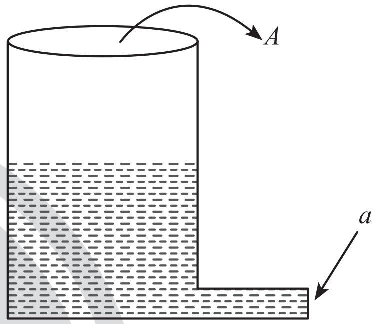
12. When a ball is dropped into a lake from a height 4.9 m above the water level, it hits the water with a velocity v and then sinks to the bottom with the constant velocity v. It reaches the bottom of the lake 4.0 s after it is dropped. The approximate depth of the lake is:

[27 June, 2022 (Shift-II)]

(*a*) 19.6 m (*b*) 29.4 m (*c*) 39.2 m (*d*) 73.5 m

13. A light cylindrical vessel is kept on a horizontal surface. Area of base is A. A hole of cross-sectional area 'a' is made just at its bottom side. The minimum coefficient of friction necessary to prevent sliding the vessel due to the impact force of the emerging liquid is ($a << A$):

[27 July, 2021 (Shift-I)]



(*a*) None of these (*b*) $\frac{2a}{A}$

(*c*) $\frac{A}{2a}$ (*d*) $\frac{a}{A}$

14. A drop of liquid of density ρ is floating half immersed in a liquid of density σ and surface tension 7.5×10^{-4} Ncm^{-1}. The radius of drop in cm will be: (g = 10 ms^{-2})

[25 July, 2022 (Shift-II)]

(*a*) $\frac{15}{\sqrt{2\rho-\sigma}}$ (*b*) $\frac{15}{\sqrt{\rho-\sigma}}$

(*c*) $\frac{3}{2\sqrt{\rho-\sigma}}$ (*d*) $\frac{3}{20\sqrt{2\rho-\sigma}}$

15. Two small drops of mercury each of radius R coalesce to from a single large drop. The ratio of total surface energy before and after the change is: **[20 July, 2021 (Shift-II)]**

(*a*) $2^{\frac{1}{3}}:1$ (*b*) $1:2^{\frac{1}{3}}$

(*c*) 2 : 1 (*d*) 1 : 2

16. In an experiment to verify Stokes law, a small spherical ball of radius r and density ρ falls under gravity through a distance h in air before entering a tank of water. If the terminal velocity of the ball inside water is same as its velocity just before entering the water surface, then the value of h is proportional to (ignore viscosity of air)

[5 Sep, 2020 (Shift-II)]

(*a*) r^4 (*b*) r^3 (*c*) r (*d*) r^2

17. When two soap bubbles of radii a and b $(b > a)$ coalesce, the radius of curvature of common surface is: **[17 March, 2021 (Shift-I)]**

(*a*) $\dfrac{ab}{a+b}$ (*b*) $\dfrac{ab}{b-a}$

(*c*) $\dfrac{b-a}{ab}$ (*d*) $\dfrac{a+b}{ab}$

18. Eight equal drops of water are falling through air with a steady speed of 10 cm/s. If the drops coalesce, the new velocity is: **[11 April, 2023 (Shift-II)]**

(*a*) 10 cm/s (*b*) 40 cm/s (*c*) 16 cm/s (*d*) 5 cm/s

19. The terminal velocity (v_t) of the spherical rain drop depends on the radius (r) of the spherical rain drop as: **[25 June, 2022 (Shift-I)]**

(*a*) $r^{1/2}$ (*b*) r (*c*) r^2 (*d*) r^3

20. A water drop of radius 1 μm falls in a situation where the effect of buoyant force is negligible. Co-efficient of viscosity of air is 1.8×10^{-5} Nsm^{-2} and its density is negligible as compared to that of water 10^6 gm^{-3}. Terminal velocity of the water drop is: **[28 June, 2022 (Shift-II)]** (Take acceleration due to gravity = 10 ms^{-2})

(*a*) 145.4×10^{-6} ms^{-1} (*b*) 118.0×10^{-6} ms^{-1}

(*c*) 132.6×10^{-6} ms^{-1} (*d*) 123.4×10^{-6} ms^{-1}

21. A raindrop with radius R = 0.2 mm falls from a cloud at a height h = 2000 m above the ground. Assume that the drop is spherical through its fall and the force of buoyance may be neglected, then the terminal speed attained by the raindrop is : **[27 July, 2021 (Shift-II)]**

[Density of water f_w = 1000 kgm^{-3} and Density of air f_a = 1.2 kg^{-3}, g = 10 m/s^2

Coefficient of viscosity of air = 1.8×10^{-5} Nsm^{-2}]

(*a*) 4.94 ms^{-1} (*b*) 14.4 ms^{-1}

(*c*) 43.56 ms^{-1} (*d*) 250.6 ms^{-1}

22. Water from a pipe is coming at a rate of 100 litres per minute. If the radius of the pipe is 5 cm, the Reynolds number for the flow is of the order of : (density of water = 1000 kg/m^3, coefficient of viscosity of water = 1mPas) **[8 April, 2019 (Shift-I)]**

(*a*) 10^6 (*b*) 10^3

(*c*) 10^4 (*d*) 10^2

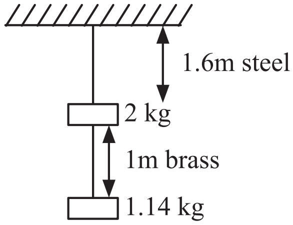
23. A solid sphere, of radius R acquires a terminal velocity v_1 when falling (due to gravity) through a viscous fluid having a coefficient of viscosity η. The sphere is broken into 27 identical solid spheres. If each of these spheres acquires a terminal velocity, v_2, when falling through the same fluid, the ratio (v_1/v_2) equals : **[12 April, 2019 (Shift-II)]**

(*a*) 27 (*b*) 1/27

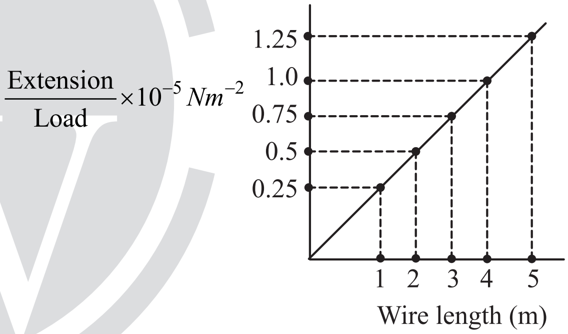
(*c*) 9 (*d*) 1/9

Integer Type Questions

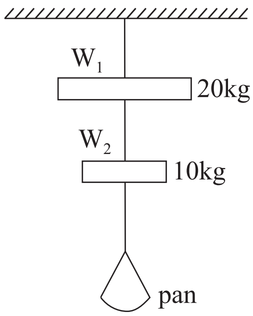
24. Two wires each of radius 0.2 cm and negligible mass, one made of steel and other made of brass are loaded as shown in the figure. The elongation of the steel wire is ______ $\times 10^{-6}$ m. [Young's modulus for steel = 2×10^{11} Nm^{-2} and g = 10 ms^{-2}] [10 April, 2023 (Shift-I)]



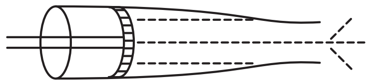
25. In an experiment to determine the Young's modulus, steel wires of five different lengths (1, 2, 3, 4, and 5m) but of same cross section (2 mm^2) were taken and curves between extension and load were obtained. The slope (extension / load) of the curves were plotted with the wire length and the following graph is obtained. If the Young's modulus of given steel wires is $x \times 10^{11}$ Nm^{-2}, then the value of x is _________. [27 July, 2022 (Shift-II)]



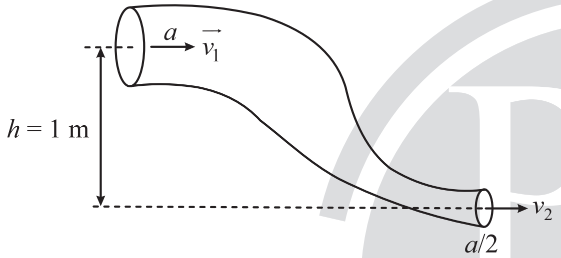
26. Wires W_1 and W_2 are made of same material having the breaking stress of 1.25×10^9 N/m^2. W_1 and W_2 have cross-sectional area of 8×10^{-7} m^2 and 4×10^{-7} m^2, respectively. Masses of 20 kg and 10 kg hang from them as shown in the figure. The maximum mass that can be placed in the pan without breaking the wires is ____ kg. (Use g = 10 m/s^2) **[27 Aug, 2021 (Shift-II)]**



27. Figure below shows a liquid being pushed out of the tube by a piston having area of cross section 2.0 cm^2. The area of cross section at the outlet is 10 mm^2. If the piston is pushed at a speed of 4 cm s^{-1}, the speed of outgoing fluid is ________ cm s^{-1}. [10 April, 2023 (Shift-II)]



28. An ideal fluid of density 800 kgm^{-3}, flows smoothly through a bent pipe (as shown in figure) that tapers in cross-sectional area from a to $a/2$. The pressure difference between the wide and narrow sections of pipe is 4100 Pa. At wider section, the velocity of fluid is $\frac{\sqrt{x}}{6}$ ms^{-1} for x =_____. (Given $g = 10$ ms^{-2}) [26 June, 2022 (Shift-I)]



29. The velocity of upper layer of water in a river is 36 kmh^{-1}. Shearing stress between horizontal layers of water is 10^{-3} Nm^{-2}. Depth of the river is ______ m. (Co-efficient of viscosity of water is 10^{-2} Pa.s)**[25 June, 2022 (Shift-I)]**

ANSWER KEY

1. (*c*)	**2.** (*d*)	**3.** (*c*)	**4.** (*c*)	**5.** (*b*)	**6.** (*c*)	**7.** (*d*)	**8.** (*d*)	**9.** (*b*)	**10.** (*b*)
11. (*a*)	**12.** (*d*)	**13.** (*b*)	**14.** (*a*)	**15.** (*a*)	**16.** (*a*)	**17.** (*b*)	**18.** (*b*)	**19.** (*c*)	**20.** (*d*)
21. (*a*)	**22.** (*c*)	**23.** (*c*)	**24.** [20]	**25.** [2]	**26.** [40]	**27.** [80]	**28.** [363]	**29.** [100]	

EXPLANATIONS

1. (*c*) Tension $(F) = mg$

$$= 4 \times \frac{10}{4} = 10\,N$$

Elongation produced in the wire is,

$$\Delta L = \frac{FL}{AY}$$

$$= \frac{10 \times 6}{3 \times 10^{-6} \times 2 \times 10^{11}} = 0.1 mm$$

2. (*d*) $Y = \dfrac{F/A}{\dfrac{\Delta\ell}{\ell}}$

$$\Rightarrow \mathrm{F} = \frac{\mathrm{YA}}{\ell}\Delta\ell$$

$$\left(\frac{\mathrm{A}\Delta\ell}{\ell}\right)_1 = \left(\frac{\mathrm{A}\Delta\ell}{\ell}\right)_2$$

$$\Rightarrow \frac{\Delta\ell_2}{\Delta\ell_1} = \frac{\mathrm{A}_1}{\mathrm{A}_2} \times \frac{\ell_2}{\ell_1}$$

$$\Rightarrow \frac{\Delta\ell_2}{0.2} = \frac{1}{2.4 \times 2.4} \times \frac{2}{1}$$

$\Rightarrow \Delta\ell^2 = 6.9 \times 10^{-2}$ mm

3. (*c*) By using Hooke's Law

$F \,\alpha\, \Delta L$

$$\frac{F_1}{F_2} = \frac{\Delta L_1}{\Delta L_2}$$

$$\Rightarrow \frac{10}{20} = \frac{L_1 - L}{L_2 - L}$$

$$\Rightarrow L = 2L_1 - L_2$$

4. (*c*) Given that,

Length of the metal wire is ℓ_1 when the tension is T_1

Length of the metal wire is ℓ_2 when the tension is T_2

Using Hooke's law,

$T_1 \propto (\ell_1 - \ell)$, and $T_2 \propto (\ell_2 - \ell)$

$$\therefore \frac{T_1}{T_2} = \frac{\ell_1 - \ell}{\ell_2 - \ell} \Rightarrow \ell = \frac{T_2\ell_1 - T_1\ell_2}{T_2 - T_1}$$

5. (*b*) Given, $\dfrac{Y_A}{Y_B} = \dfrac{7}{4}$

$L_A = 2$ m $A_A = \pi R^2$

$L_B = 1.5$ m $A_B = \pi$ (2 mm)^2

$$\frac{F}{A} = Y\left(\frac{\ell}{L}\right)$$

Since F and ℓ are same $\Rightarrow \dfrac{AY}{L}$ is same

$$\frac{A_A Y_A}{L_A} = \frac{A_B Y_B}{L_B} \Rightarrow \frac{(\pi R^2)\left(\frac{7}{4}Y_B\right)}{2} = \frac{\pi(2\text{ mm})^2 .Y_B}{1.5}$$

$R = 1.74$ mm

6. (*c*) As length is same

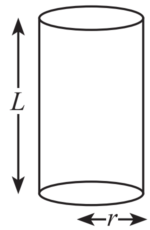
$$\text{So } \left(\frac{F}{A}\right)_{\text{Compressive}} = \left(\frac{F}{A}\right)_{\text{Thermal}}$$

$$\frac{F}{\pi r^2} = Y\alpha T$$

linear coefficient of expansion $\alpha = \dfrac{F}{YT\pi r^2}$

$\therefore$ The coefficient of volume expansion $\gamma = 3\alpha$

$$\therefore \gamma = 3\frac{F}{YT\pi r^2}$$



7. (*d*) We know that $K = -V\dfrac{dp}{dv}$

Also as mass remains constant we have $\dfrac{dv}{v} = \dfrac{d\rho}{\rho}$

This we get, $K = -\rho\dfrac{dp}{d\rho}$

Hence, $dp = p$

So, $d\rho = -\dfrac{\rho p}{K}$

8. (*d*) $\dfrac{du}{dv} = \dfrac{1}{2}\text{stress}\times\dfrac{\text{stress}}{y} = \dfrac{1}{2}\dfrac{F^2}{A^2 y}$

$$\frac{du}{dv} \propto \frac{1}{d^4}$$

$$\frac{\left(\dfrac{du}{dv}\right)_1}{\left(\dfrac{du}{dv}\right)_2} = \frac{d_2^4}{d_1^4} = \frac{1}{4}$$

$$\frac{d_1}{d_2} = (4)^{1/4}\ ;\ \frac{d_1}{d_2} = \sqrt{2}:1$$

9. (*b*) $h_1 + h_2 = 0.29 \times 2 + 0.1$

$h_1 + h_2 = 0.68$ ...(*a*)

$\Rightarrow P_0 + \rho_k g(0.1) + \rho_w g(h_1 - 0.1)$

[ρ_k = density of kerosene and ρ_w = density of water]

$-\rho_w g h_2 = P_0$

$\Rightarrow\ \rho_k g(0.1) + \rho_w g h_1 - \rho_w g \times (0.1) = \rho_w g h_2$

$\Rightarrow\ 800 \times 10 \times 0.1 + 1000 \times 10 \times h_1$

$-1000 \times 10 \times 0.1 = 1000 \times 10 \times h_2$

$\Rightarrow\ 10000\,(h_1 - h_2) = 200$

$\Rightarrow\ h_1 - h_2 = 0.02$ ...(*b*)

$\Rightarrow\ h_1 = 0.35$

$\Rightarrow\ h_2 = 0.33$

So, $\dfrac{h_1}{h_2} = \dfrac{35}{33}$

10. (*b*) $dVg = \rho\left(\dfrac{V}{2}\right)g + T(2\pi r)$

$$\Rightarrow d.\frac{4}{3}\pi r^3 g = \rho.\frac{2}{3}\pi r^3 g + 2\pi r T$$

$$\Rightarrow \frac{2}{3}r^2 g(2d - \rho) = 2T$$

$$\Rightarrow r = \sqrt{\frac{3T}{(2d-\rho)g}}$$

11. (*a*) From continuity equation $A_A V_A = A_B V_B$

$1.5 \times V_A = 25 \times 10^{-2} \times 60$

$$V_A = \frac{25 \times 60 \times 10^{-2} \times 10}{1.5}$$

$V_A = 10$ cm/s

By Bernoulli's theorem

$$P_A + \frac{1}{2}\times 1000 \times (0.1)^2 = P_B + \frac{1}{2}\times 1000 \times (0.6)^2$$

$$P_A + 5 = P_B + \frac{1}{2}\times 1000 \times 36 \times 10^{-2}$$

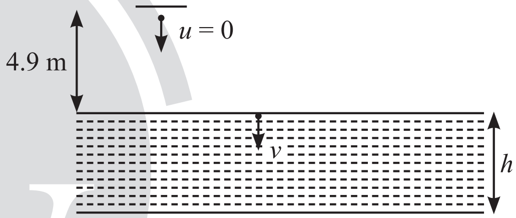
$P_A + 5 = P_B + 180$

$P_A - P_B = 175$ Pa

12. (*d*) Given,

Total time, $T = 4$ sec

$u = 0$ m/sec, $g = 9.8$ m/sec^2, $h = 4.9$ m, $t = ?$



Let time taken by ball to hit the surface of water be t, Therefore, from kinematic equation,

$$\Rightarrow h = ut + \frac{1}{2}gt^2$$

$$4.9 = 0.t + \frac{1}{2}\times 9.8t^2$$

$t = 1$ s

Now, let assume 'v' be the velocity with which ball hits the water, therefore

$v = u + at$

$= 0 + 9.8 \times 1 = 9.8$ m/sec

Time taken to reach the bottom of the lake from surface of the lake

$= 4 - 1 = 3$ sec

$v = 9.8$ m/sec

Again, from kinematic equation

$$H = 9.8 \times 3 + \frac{1}{2}\times 9.8 \times (3)^2$$

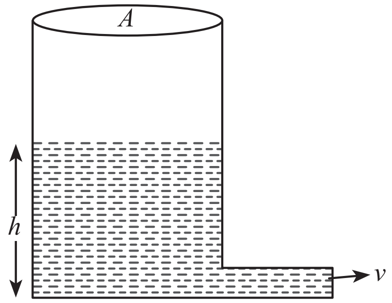
$H = 73.5$ m

13. (*b*) Velocity of efflux, $v=\sqrt{2gh}$

Thrust on the hole

$$F=v\frac{dm}{dt}=v(a\rho v)=v^2a\rho=2gha\rho$$

[$\because$ dm = Area × height × density]



Normal reaction on the hole, $R = mg = AH\rho g$

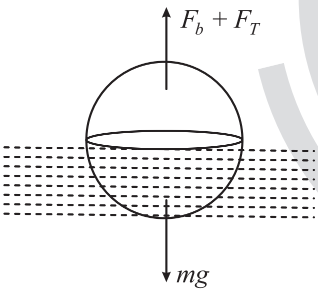
Therefore, friction force $f_r = \mu R = \mu Ah\rho g$

For just sliding, $f_r = F$

$\mu Ah\rho g = 2a\rho gh$

or $\mu=\dfrac{2a}{A}$

14. (*a*)



Buoyant force + surface tension = mg

$$\sigma\frac{V}{2}g+2\pi RT=\rho Vg$$

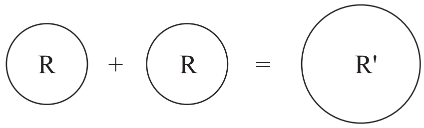
$$2\pi RT=\frac{(2\rho-\sigma)}{2}\frac{4}{3}\pi R^3\,g; \qquad \left[V=\frac{4}{3}\pi R^3\right]$$

$$R^3=\frac{3T}{(2\rho-\sigma)g}\Rightarrow R=\sqrt{\frac{3\times7.5\times10^{-2}\,N-m^{-1}}{(2\rho-\sigma)\times10}}$$

$$R=\frac{3}{20\sqrt{(2\rho-\sigma)}}\text{m}=\frac{15}{\sqrt{2\rho-\sigma}}\text{cm}$$

15. (*a*) Total surface energy before coalesce = $U_i = 2\times4\pi R^2\times\sigma = 8\pi R^2\sigma$ ...(*i*)

Let new radius becomes r, so according to conservation energy we can write



$$2\times\frac{4\pi R^3}{3}=\frac{4\pi r^3}{3}\Rightarrow r=2^{\frac{1}{3}}R$$

Total surface energy after coalesce

$$=U_f=4\pi r^2\times\sigma=2^{\frac{2}{3}}\times4\pi R^2\sigma \qquad ...(ii)$$

From equation (*i*) and (*ii*)

$$\Rightarrow\frac{U_i}{U_f}=\frac{8\pi R^2\sigma}{2^{\frac{2}{3}}\times4\pi R^2\sigma}=2^{\frac{1}{3}}:1$$

16. (*a*) $v_T \propto r^2$

$\sqrt{2gh}\propto r^2$

$\Rightarrow\ h\propto r^4$

17. (*b*) $P_1-P_0=\dfrac{4T}{a}$

$P_2-P_0=\dfrac{4T}{b}$

Resultant of excess pressure inside soap bubble,

$$p=P_1-P_2=4T\left(\frac{1}{a}-\frac{1}{b}\right)$$

$$=\frac{4T}{r}=4T\left(\frac{1}{a}-\frac{1}{b}\right)\Rightarrow r=\frac{ab}{b-a}$$

18. (*b*) From $v=\dfrac{2r^2}{g}\dfrac{(\rho-\delta)g}{\eta}$,

Let the radius of small drop be r_1 and bigger drop be r_2.

$$\frac{v_1}{v_2}=\left(\frac{r_1}{r_2}\right)^2$$

Volume of 8 equal drops = Volume of new drop

$$8\times\frac{4}{3}\pi r_1^3=\frac{4}{3}\pi r_2^3$$

$r_2 = 2r_1$

$$\frac{10}{v_2}=\left(\frac{1}{2}\right)^2$$

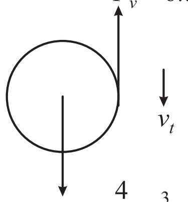
$v_2 = 40$ cm/s

19. (*c*) Formula of terminal velocity

$$v_T = \frac{2}{9}\frac{r^2(\rho-\delta)g}{\eta}$$

$$v_T \propto r^2$$

20. (*d*) $F_v = 6\pi\eta r v_t$



$$mg = \frac{4}{3}\pi r^3 \rho g$$

$$6\pi\eta r v_t = \frac{4}{3}\pi r^3 \rho g$$

$$v_t = \frac{2}{9}\times\frac{\rho r^2 g}{\eta}$$

$$= \frac{2\times10^{-12}\times10^3\times10}{9\times1.8\times10^{-5}} = 123.4\times10^{-6}\ m/s$$

21. (*a*) At terminal speed

$$Mg = Fv = 6\pi\eta R v$$

$$\Rightarrow v = \frac{mg}{6\pi\eta R}$$

$$v = \frac{\delta\times\frac{4}{3}\pi R^3 g}{6\pi\eta R}$$

$$= \frac{2\times\delta R^3\times g}{9\eta}$$

$$= \frac{2\times1000\times\left(0.2\times10^{-3}\right)^3\times10}{9\times1.8\times10^{-5}} = \frac{400}{81} = 4.94 m/s$$

22. (*c*) Reynolds Number $= \frac{\rho v d}{\eta}$

Volume flow rate = $v \times \pi r^2$

$$v = \frac{100\times10^{-3}}{60}\times\frac{1}{\pi\times25\times10^{-4}}$$

$$v = \frac{2}{3\pi}\ \text{m/s}$$

Reynolds Number $= \frac{10^3\times2\times10\times10^{-2}}{10^{-3}\times3\pi} \cong 2\times10^4$

order 10^4

23. (*c*) As mass of the sphere is same

$$\therefore \rho\frac{4}{3}\pi R^3 = 27\cdot\frac{4}{3}\pi r^3\rho$$

$$\Rightarrow r = \frac{R}{3}$$

Now, $V_T = \frac{2}{9}\frac{r^2}{\eta}(\rho_0-\rho_\ell)g$

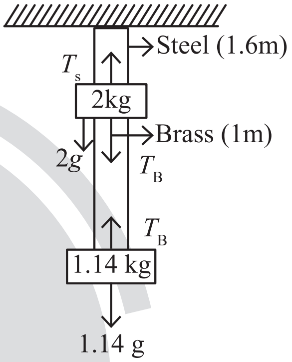
$$\Rightarrow V_T \propto r^2$$

$$\therefore \frac{V_1}{V_2} = \frac{R^2}{r^2} = \frac{R^2}{\frac{R^2}{9}} = \frac{9}{1}$$

24. [20] Tension in steel wire

$T_S = 2g + T_B$

$T_S = 20 + 11.4 = 31.4\ N$



For steel wire $\Delta L = \frac{T_S L}{Ay}$

$$\Delta L = \frac{31.4\times1.6}{\pi\left(0.2\times10^{-2}\right)^2\times2\times10^{11}} = 20\times10^{-6}\ m$$

25. [2] Slope of the curve $= \frac{\Delta L/W}{L} = \frac{\Delta L/L}{W} = \frac{1}{YA}$

$$Y = \frac{1}{(\text{Slope})\times A}$$

$$Y = \frac{1}{\left(0.25\times10^{-5}\right)\times\left(2\times10^{-6}\right)}$$

$$Y = 2\times10^{11}\ N/m^2$$

26. [40] Given,

Cross-sectional area of $W_1 = 8\times10^{-7}\ \text{m}^2$

Cross-sectional area of $W_2 = 4\times10^{-7}\ \text{m}^2$

For W_1,

Breaking stress = $1.25\times10^9\ \text{N/m}^2$

$$\frac{(m+10+20)\times10}{8\times10^{-7}} \le 1.25\times10^9$$

$$\Rightarrow m \le 70\text{kg}$$

For W_2,

Breaking stress = 1.25 × 109 N/m²

$$\frac{(m+10)\times 10}{4\times 10^{-7}} \le 1.25\times 10^{9}$$

$\Rightarrow m \le 40\text{kg}$

Thus, $m \le 40\text{kg}$

27. [80] Using equation of continuity

$$A_1V_1 = A_2V_2$$

$$V_2 = \frac{A_1V_1}{A_2} = \frac{2\times 4}{10\times 10^{-2}} = 80\,cm/s$$

28. [363] From continuity equation $av_1 = \frac{a}{2}v_2$

$$v_2 = 2v_1$$

From Bernoulli's theorem,

$$P_1 + \rho g h_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g h_2 + \frac{1}{2}\rho v_2^2$$

$$4100 = 800\left[\left(\frac{4v_1^2 - v_1^2}{2}\right) + 10\times(0-1)\right]$$

$$\frac{41}{8} + 10 = \frac{3v_1^2}{2}$$

$$v_1 = \sqrt{\frac{121}{4\times 3}\times\frac{3}{3}}$$

$$v_1 = \frac{\sqrt{363}}{6}\text{m/s}$$

$X = 363$

29.[100] $F = \eta A\frac{\Delta v_x}{\Delta y}$

$$\frac{F}{A} = \eta\frac{\Delta v_x}{\Delta y}$$

$$\Rightarrow 10^{-3} = 10^{-2}\times\frac{36\times 1000}{h\times 3600}$$

$$\Rightarrow h = 10^{-2}\times\frac{36\times 1000}{10^{-3}\times 3600} = 100\,\text{m}$$